Combinatorics 2

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Problem 1

Let $N = p_1 p_2 \dots p_n$. Splitting N into its prime factors is the same as splitting the set $\{p_1, \dots, p_n\}$ into nonempty subsets, which is $B_n = \sum_{k=0}^{n} \left\{ {n \atop k} \right\}$.

Problem 2

To create a permutation of $[n]$ with exactly 2 descents, we can split $[n]$ into three disjoint sets, A, B, C, and write each set's elements side by side in ascending order. However, this sometimes creates 1 or 0 descents, so we exclude those from our total count.

The number of ways to split $[n]$ into three subsets is 3^n. However, if every element in A is less than every element in B, whose elements are less than every element in C, the permutation does not have 2 descents. The number of these permutations is $(n+1)^2$, since we choose two points between 0 and n to be the splitting point between A, B and B, C.

Also, if every element in A is less than every element in B, the permutation doesn't have 2 descents. We can count the number of these sets by fixing the size of C to be k. Then there are $n-k+1$ ways to choose the split point between A and B and $\binom{n}{k}$ ways to choose C, so the total count is

$$\sum_{k=0}^{n} \binom{n}{k}(n-k+1) = \sum_{k=0}^{n} \binom{n}{k} n - \sum_{k=0}^{n} \binom{n}{k} k + \sum_{k=0}^{n} \binom{n}{k}$$

$$= (n+1) \sum_{k=0}^{n} \binom{n}{k} - n \cdot 2^{n-1} = (n+1) \cdot 2^n - n \cdot 2^{n-1}$$

$$= (n+2) \cdot 2^{n-1}.$$

This is also the number of ways to make the split where every element in B is less than every element in C. However, summing both cases overcounts the case where all of A is less than all of B which is also less than all of C, so we

subtract that count once.

The overall sum is

$$\begin{aligned} &3^n - (2 \cdot (n+2) \cdot 2^{n-1} - (n+1)^2) \\ &= 3^n - 2^n(n+2) + (n+1)^2. \end{aligned}$$

Problem 3

For the lower bound, we can make a set partition of $[2n]$ into $[n]$ blocks as follows. First, start with the blocks $\{1\}, \{2\}, \ldots, \{n\}$. Then, for each element $n+1, \ldots, 2n$, choose one of the n blocks to drop it into. Then the resulting partition contains n nonempty blocks, so it is counted by $\left\{ {2n \atop n} \right\}$. The number of ways to do this is n^n, so for $n \geq 2$,

$$\left\{ {2n \atop n} \right\} \geq n^n > n!.$$

For the upper bound, we form a permutation of $[2n]$ given a set partition of $[2n]$ into n blocks. First, sort each block internally in increasing order, and then order all the blocks by their minimum element. Finally, concatenate the blocks to form a permutation of $[2n]$. Each set partition produces a unique permutation, so $\left\{ {2n \atop n} \right\} \leq (2n)!$. Furthermore, the inequality is strict for $n \geq 2$, since a permutation starting with $2, 1, 3, 4, \ldots$ can't be produced in this way.

Problem 4

The number of set partitions of $[n]$ with $\{1\}$ as one of the subsets is B_{n-1}, so the number of set partitions of $[n]$ without $\{1\}$ is $B_n - B_{n-1}$.

Problem 5

First, we count the number of ways to split $[n]$ into two parts, which is $\left\{ {n \atop 2} \right\}$, and subtract the number of ways to split $[n]$ into two subsets of equal size, which is $\binom{n}{n/2}$. So the result is $\left\{ {n \atop 2} \right\} - \binom{n}{n/2}$.

Problem 6

Let a_n be the number of permutations that have 1 and 2 in the same cycle. Given a permutation of $[n-1]$ with 1 and 2 in the same cycle, we can form a

permutation of $[n]$ by placing n in front of any of the existing numbers, or we can place it in its own cycle. So $a_n = (n-1)a_{n-1} + a_{n-1} = na_{n-1}$ for all $n \geq 3$. We have that $a_1 = 0$ and $a_2 = 1$, so $a_3 = 3$, $a_4 = 12$, and so on. In general, $a_n = n!/2$.

Let b_n be the number of permutations that have 1, 2, and 3 in the same cycle. Similar to before, $b_n = nb_{n-1}$ for $n \geq 4$. We have that $b_1 = b_2 = 0$ and $b_3 = 3!/3 = 2$, so $b_4 = 8$, $b_5 = 40$, and in general, $b_n = n!/3$.

Let c_n be the number of permutations with 1 and 2 in the same cycle and 3 and 4 in the same cycle. We have that $c_n = nc_{n-1}$ for all $n \geq 5$. Also, $c_1 = c_2 = c_3 = 0$, and $c_4 = 1 + 4!/4 = 7$. So $c_n = n(n-1)\cdots(6)(5)\cdot 7 = n! \cdot \frac{7}{24}$.

Problem 7

Assume the n people sit at i tables. Then the number of ways to seat them is $\begin{bmatrix} n \\ i \end{bmatrix}$ and the number of ways for all of them to choose a menu item is k^i. Summing over i, we get a total of

$$\sum_{i=0}^{k} \begin{bmatrix} n \\ i \end{bmatrix} k^i.$$

Problem 8

Take any set partition of $[n]$ (this can be done in B_n ways). If the partition has no blocks of size 1, keep it as is (counted by a_n). Otherwise, merge all the blocks of size 1 into one block and add $n+1$ (counted by a_{n+1}). This process is reversible, since we can take the block with $n+1$ and split its elements into individual blocks. So we have a bijection, and $B_n = a_n + a_{n+1}$.

Problem 15

Part (a)

The LHS is the number of ways to split $[n+1]$ into $m+1$ disjoint nonempty subsets. We can also count this by the size of the block that contains $n+1$. We can choose k elements that are not in that block in $\binom{n}{k}$ ways and partition the remaining k elements in $\left\{ {k \atop m} \right\}$ ways. Summing over k gives us the RHS.

Part (b)

The LHS is the number of permutations of $[n+1]$ with $m+1$ cycles. To count this, we can start with a permutation of $[n]$ with k cycles, and choose m of them

to be left as separate cycles. We can concatenate the other $k - m$ cycles into one big cycle along with the new element $n+1$. This is reversible, since we can delete $n+1$ and split its cycle into the k original cycles. So the LHS and RHS are equal.

Problem 17

Each card in a deck can be mapped to a set partition of [5]. I was thinking of something where we ask the spectator to secretly split five different objects into a few groups, then we ask a few identifying questions (as seemingly unrelated as possible) to isolate a card, and then map that card to the spectator's set partition. However I couldn't come up with anything more specific.

Problem 18

Part (a)

Assume $f(x) = x^m$. Simplifying the LHS, we get

$$x\frac{d}{dx}(x^m) = x(mx^{m-1}) = mx^m,$$

$$\left(x\frac{d}{dx}\right)^n (x^m) = m^n x^m.$$

Simplifying the RHS,

$$x^k \frac{d^k}{dx^k}(x^m) = m(m-1)\cdots(m-k+1)x^m = m^{\underline{k}} x^m,$$

$$\sum_{k=0}^{n} \left\{ n \atop k \right\} x^k \frac{d^k}{dx^k}(x^m) = \sum_{k=0}^{n} \left\{ n \atop k \right\} m^{\underline{k}} x^m = m^n x^m.$$

So the LHS and RHS are equal. Since summations and the derivative operator are linear, this applies to all polynomial functions f.

Part (b)

Again, assume $f(x) = x^m$. The LHS becomes

$$x^n \frac{d^n}{dx^n}(x^m) = m^{\underline{n}} x^m.$$

The RHS becomes

$$\sum_{k=0}^{n} (-1)^{n-k} \left[n \atop k \right] x^k \frac{d^k}{dx^k}(x^m) = \sum_{k=0}^{n} (-1)^{n-k} \left[n \atop k \right] m^k x^m = m^{\underline{n}} x^m.$$

So the LHS and RHS are equal for all polynomial functions f.